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COURSE CODE: SBT-DF203
LAB NAME: BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS
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INSTRUCTOR: AMINU IDRIS

Scenario

Your practical context

Act as a junior network forensic analyst and document a controlled plaintext HTTP session. Create a local training webpage, capture the traffic, reconstruct the TCP conversation, identify the HTTP request and response, and explain the evidential value of HTTP, TCP, IP and link-layer metadata.

Objectives

What you must demonstrate

Explain HTTP/TCP/IP forensic artefacts; capture a complete HTTP session; identify SYN, SYN-ACK and ACK; extract ports, sequence and acknowledgement numbers, timestamps and HTTP headers; reconstruct the session; document integrity hashes and a concise forensic timeline.

Instructor briefing

Laboratory guidance

SBT-DF203 — Basic Networking Skills for Digital Forensics

Lab 1 — HTTP Analysis Using Wireshark: Text Traffic

Official Lab Manual

Read the complete manual before starting. [Open the official SBT-DF203 Lab 1 manual](#).

The manual contains the detailed commands, evidence worksheets, marking rubric, screenshot checklist, safety controls and report structure. The portal instructions below are the assessed workflow summary and do not replace the manual.

Required Practical Workflow

1. Read the complete official lab manual before running any command.
2. Create the required evidence, working, reports and screenshots folders and record the case/lab identifier.
3. Create the authorised local Apache webpage containing your name and registration number and verify TCP port 80.
4. Capture the HTTP session with Wireshark or TShark, preserve the original PCAP/PCAPNG and calculate SHA-256 hashes.

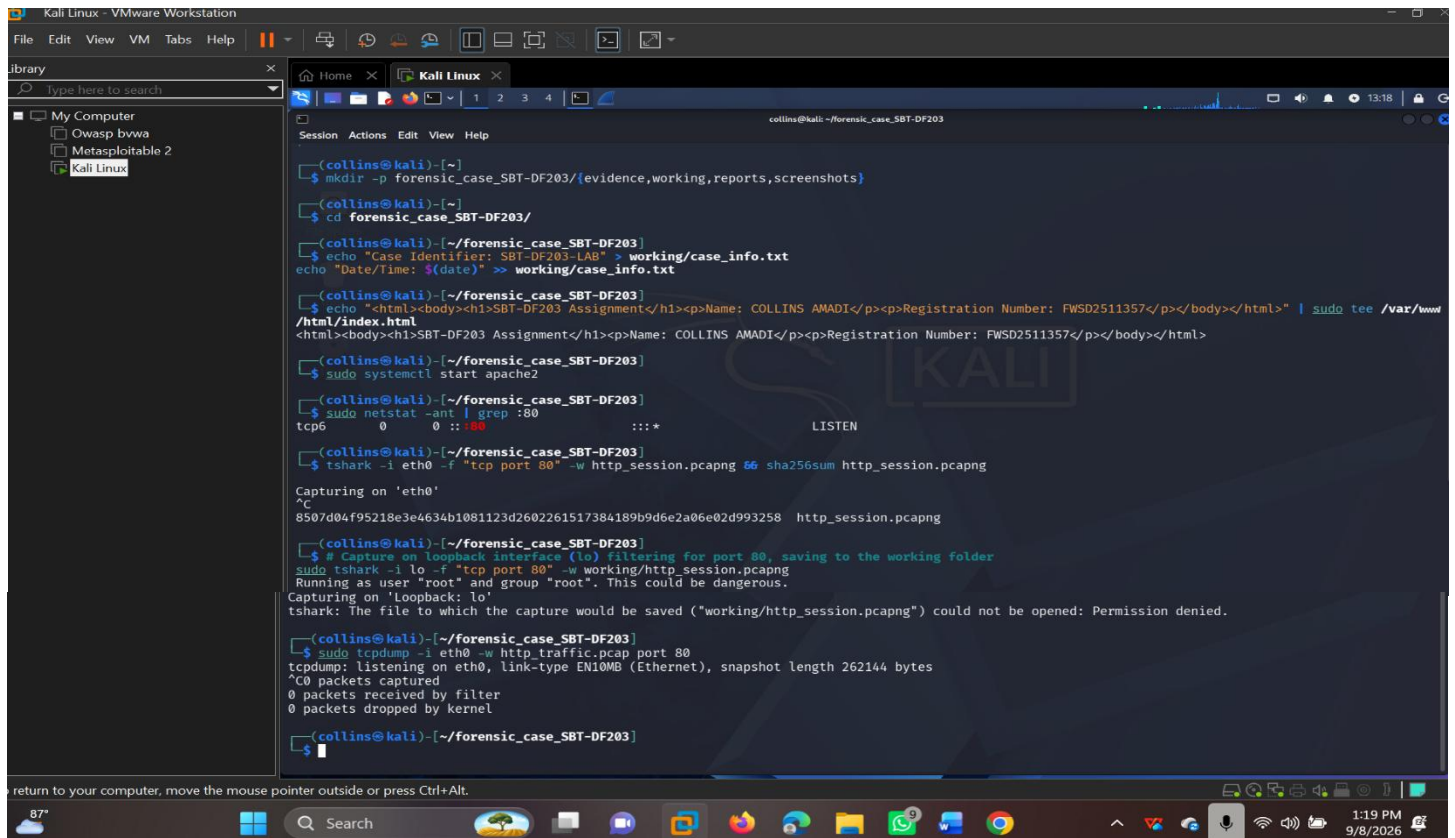
5. Analyse the TCP three-way handshake, HTTP GET request, HTTP response, encapsulation and connection termination.
6. Prepare the required findings table, screenshots and professional report.

Authorised Training Resource

Optional basic HTTP training capture

Use this resource only for the SBT-DF203 authorised training exercise. Preserve the source file and work from a verified copy where the manual requires it.

ANSWER:



```
(collins@kali)-[~]
$ mkdir -p forensic_case_SBT-DF203/{evidence,working,reports,screenshots}
(collins@kali)-[~]
$ cd forensic_case_SBT-DF203/
(collins@kali)-[~/forensic_case_SBT-DF203]
$ echo "Case Identifier: SBT-DF203-LAB" > working/case_info.txt
echo "Date/Time: $(date)" >> working/case_info.txt
(collins@kali)-[~/forensic_case_SBT-DF203]
$ echo "<html><body><h1>SBT-DF203 Assignment</h1><p>Name: COLLINS AMADI</p><p>Registration Number: FWSD2511357</p></body></html>" | sudo tee /var/www/html/index.html
<html><body><h1>SBT-DF203 Assignment</h1><p>Name: COLLINS AMADI</p><p>Registration Number: FWSD2511357</p></body></html>
(collins@kali)-[~/forensic_case_SBT-DF203]
$ sudo systemctl start apache2
(collins@kali)-[~/forensic_case_SBT-DF203]
$ sudo netstat -ant | grep :80
tcp6      0      0  :::*          :::*          LISTEN
(collins@kali)-[~/forensic_case_SBT-DF203]
$ tshark -i eth0 -f "tcp port 80" -w http_session.pcapng 66 sha256sum http_session.pcapng
Capturing on 'eth0'
8507d04f95218e3e4634b1081123d2602261517384189b9d6e2a06e02d993258 http_session.pcapng
(collins@kali)-[~/forensic_case_SBT-DF203]
$ # Capture on loopback interface (lo) filtering for port 80, saving to the working folder
sudo tshark -i lo -f "tcp port 80" -w working/http_session.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
tshark: The file to which the capture would be saved ("working/http_session.pcapng") could not be opened: Permission denied.
(collins@kali)-[~/forensic_case_SBT-DF203]
$ sudo tcpdump -i eth0 -w http_traffic.pcap port 80
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
(collins@kali)-[~/forensic_case_SBT-DF203]
$
```

